

Multi-Agent Reinforcement Learning Based UAV Swarm Communications Against Jamming

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Abstract—The swarm relay and power allocation policy determines the bit error rate and the energy consumption of unmanned aerial vehicles (UAVs) and can be optimized based on the network and jamming model, which is rarely known by UAVs. In this paper, we propose a multi-agent reinforcement learning (RL)-based UAV swarm communication scheme to optimize the relay selection and power allocation against jamming. Based on the network topology, channel states, previous performance and observations shared by the neighboring UAVs, this scheme formulates the policy distribution to improve the policy exploration and applies a policy learning mechanism to stabilize the learning process. Based on transfer learning, the shared swarm experiences are exploited to accelerate the initial learning and improve policy optimization. A deep RL-based scheme is proposed to mitigate the state quantization error for the rapidly changing channel states under high swarm moving speed and thus further improve the anti-jamming performance. This scheme designs a policy network with four fully connected layers to approximate the policy distribution and uses another two neural networks to estimate the average policy distribution and the expected long-term utility, respectively, to update the policy network for stabilized deep learning. We investigate the computational complexity and derive the performance bound regarding the bit error rate, the energy consumption and the utility. Simulation and experimental results verify the performance gain of our proposed schemes over related works.

Index Terms—Unmanned aerial vehicles, swarm communications, jamming, reinforcement learning.

I. INTRODUCTION

UNMANNED aerial vehicle (UAV) swarm communications against interference and jamming are critical to support applications such as remote sensing and traffic surveillance that rely on the reliable transmission of flight status, commands and sensing data [1], [2], [3], [4]. For example,

Manuscript received 28 July 2022; revised 25 November 2022; accepted 9 April 2023. Date of publication 24 April 2023; date of current version 12 December 2023. This work was supported by the National Natural Science Foundation of China under Grant 61971366 and Grant U21A20444. The associate editor coordinating the review of this article and approving it for publication was Y. Shu. (*Corresponding author: Liang Xiao.*)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TWC.2023.3268082>.

Digital Object Identifier 10.1109/TWC.2023.3268082

the swarm target tracking in [3] requires the reliable transmission of the 24-bit target position information per second. The UAV relay and power selection that has been optimized in [4] to improve the communication performance and energy efficiency under known jamming model and channel states of the relay-destination links, which are rarely known by UAVs.

This problem has been addressed by reinforcement learning (RL) in wireless communication systems [5], [6]. For example, the wireless system in [6] applies Q-learning to choose the relay node based on the network topology and channel conditions to the neighboring radio devices and applies Lagrange dual decomposition to allocate the transmit power for higher energy efficiency. However, with the sample complexity increased with the swarm size and moving speed, these schemes suffer from severe performance degradation if directly applied in large-scale dynamic swarms.

In this paper, we propose a multi-agent RL-based swarm relay selection and power allocation scheme that exploits the observations and experiences shared among the swarm to enhance the UAV perceptive field and improve the anti-jamming communication performance of large-scale dynamic swarms. The network states, e.g., the channel states and the number of neighboring UAVs, shared by the neighboring UAVs in both the request to send (RTS) and clear to send (CTS) messages are used to formulate the learning state in the policy selection. Based on the observed network topology, channel states, previous performance, and the shared states of neighboring UAVs, the policy distribution learning mechanism introduces randomness for more efficient policy exploration.

In this scheme, the expected long-term utility (i.e., Q-value) and the average policy distribution guide the policy distribution update to stabilize the learning process. The policy learning rate is chosen based on both the current expected Q-value and the previous expected Q-value for the dynamic swarm against the changing jamming policy, i.e., the learning rate increases for learning fast under high previous expected Q-value. Transfer learning is applied to exploit the experiences of neighboring UAVs in the initialization of the learning parameters such as policy distribution to improve policy exploration.

We further propose a deep RL-based scheme to mitigate the state quantization error and accelerate learning for rapidly changing network states under high moving speed and large swarm size. The policy network with four fully-connected layers is designed to estimate the policy distribution and the average policy network and Q network with the same

architecture estimates the average policy distribution and the Q-value of each state-action pair, respectively, to evaluate the policy distribution for the stable update of the policy network.

We investigate the computational complexity of the proposed schemes and formulate the swarm anti-jamming communication game between the N -UAV swarm and the jammer. In the game, the swarm chooses the relay node, transmit and relay power in each time slot to increase the signal-to-interference-plus-noise ratio (SINR) with less power while the jammer chooses the jamming power to decrease the swarm utility with less jamming power. The performance bound can be obtained based on the Nash equilibrium (NE) of the game under the specified channel conditions and the proposed schemes converge to the performance bound.

Simulation is performed with a 10-UAV swarm against a random jammer based on the line-of-sight (LoS) dominant channel of the swarm communication links and the probabilistic LoS jamming channel. Simulation results show that our proposed schemes reduce the message bit error rate (BER) with less energy consumption and increase the utility compared with the Q-learning based relay selection scheme (QRS) in [6]. The experimental results based on a 4-UAV swarm also verify the performance gain of our proposed schemes in terms of the average performance of the BER and energy consumption over the UAVs.

The rest of this paper is organized as follows. We review the related work in Section II, and present the system model in Section III. We present the multi-agent RL-based swarm communication scheme and the deep RL-based scheme in Sections IV and V, respectively. The performance bound and computational complexity are provided in Section VI. We report the simulation and experimental results in Section VII and Section VIII and draw the conclusion in Section IX.

II. RELATED WORK

The UAV performance such as the BER and energy consumption can be enhanced by the power allocation [7], [8], the trajectory design [9], [10] and the frequency hopping [11]. For example, the UAV power allocation scheme in [7] formulates a prospect theoretic game and applies win or learn fast policy hill-climbing and deep Q network to increase the secrecy capacity against smart attacks that launch both the eavesdropping and jamming attacks. The wireless sensor network in [10] that applies block coordinate descent and successive convex approximation to schedule the ground sensors and plan the UAV trajectory increases the achieved data rate of UAV against jamming. A UAV-enabled mobile edge computing (MEC) network in [8] that applies compressive sensing and one-dimensional search to optimize the power allocation, user association, computation allocation and location planning significantly improves the energy efficiency of MEC networks.

UAV relay using amplify-and-forward or decode-and-forward techniques helps the source such as mobile devices or vehicles forward the message to the destination to improve the communication coverage and anti-jamming performance by optimizing the placement, transmit power, and trajectory [12], [13], [14], [15]. For example, a reliable UAV swarm communication scheme in [12] applies a two-phase transmission

TABLE I
SUMMARY OF SYMBOLS AND NOTATIONS

Symbol	Description
N	Swarm size
$c_m^{(k)}$	Message size of UAV m at time slot k
$d_m^{(k)}$	Destination of UAV m
$p_m^{(k)} \in [P_{\min}, P_U]$	Transmit power of UAV m
$r_m^{(k)}$	Relay node of UAV m
$x_m^{(k)} \in [0, P_U]$	Relay power of UAV $r_m^{(k)}$
$z^{(k)} \in [0, P_J]$	Jamming power
$h_{m,i}^{(k)}$	Channel gain between UAV m and UAV i
$g_m^{(k)}$	Channel gain between the jammer and UAV m
$\beta^{(k)}$	Number of neighboring UAVs of UAV m
$\rho_m^{(k)} / \tilde{\rho}_m^{(k)}$	BER level at the destination/relay of UAV m
$\xi_m^{(k)} / \tilde{\xi}_m^{(k)}$	SINR of UAV m signal at the destination/relay
$b_m^{(k)}$	Battery level of UAV m

protocol to cooperatively transmit the common control message in the swarm and characterizes the reliability performance as the successful decoding rate of the control message. The policy hill climbing based UAV relay scheme in [13] chooses the relay policy based on the channel states and previous SINR to reduce the message BER against smart jamming. A wireless relay network in [15] deploys the unmanned aircraft as the resilient moving relay to help transmit data for ground stations with disconnected communication links and optimizes the rate of decode-and-forward protocol to decrease the outage probability and improve the data rate.

Relay selection and power allocation further enhance the relay-aided communication performance [4], [6], [16], [17], [18]. For example, a multiple-UAV aided downlink cellular network in [16] applies successive convex optimization method to jointly optimize the resource allocation and placement of UAVs and improve the downlink data rate. A hybrid satellite-UAV network in [4] applies a computationally efficient method and Lagrange multipliers optimization method to choose the UAV relay and power allocation and guarantee the quality of service (QoS) of terrestrial users. A multiple relay-aided communication scheme in [6] applies Q-learning and the Lagrange dual decomposition method to choose the relay node and power allocation and improve the energy efficiency.

Multi-agent RL such as multi-agent deep deterministic policy gradient (MADDPG) and multi-agent proximal policy optimization (MAPPO) enables multi-agent systems to design the trajectory and allocate resources for higher QoS with reduced communication overhead [19], [20], [21], [22], [23], [24], [25], [26], [27]. A vehicular network in [20] applies MADDPG to optimize the UAV resource allocation and vehicle association and improves the QoS satisfaction ratio. An energy-efficient wireless communication network in [22]

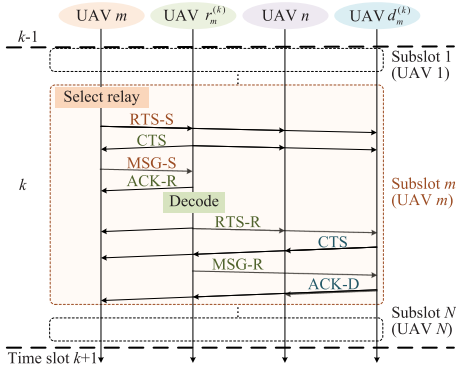


Fig. 1. Timeline of the swarm anti-jamming communications, in which UAV m aims to transmit $c_m^{(k)}$ -byte data to UAV $d_m^{(k)}$ with power $p_m^{(k)}$ and chooses UAV $r_m^{(k)}$ to relay the message with power $x_m^{(k)}$ against jamming and path loss.

applies independent Q-learning to choose the communication user, transmit power and channel and thus reduce the UAV information exchange overhead and guarantee the QoS. A cellular Internet of UAVs in [23] utilizes deep multi-agent RL to optimize the trajectory design and improve the sensing QoS. An adaptive UAV communication scheme in [24] exchanges the status information within the position packet and RTS/CTS and formulates a policy distribution to choose the routing path. In [28], we propose an RL-based UAV swarm anti-jamming communication scheme to choose the relay node and power allocation and improve communication performance. In this paper, we further propose a deep RL-based scheme to optimize the relay selection and power allocation and thus improve the learning efficiency for the highly dynamic swarm. We also derive the NE of the game, analyze the influence of the channel states and network topology on the performance bound and verify the efficacy of the proposed schemes via simulations and experiments.

III. SYSTEM MODEL

A. Network Model

The swarm equipped with sensors and cameras captures photos and surveillance videos to support traffic control applications or safety flight guidance. The sensing data and the control message are shared via TDMA within the swarm, and in each time slot, each of N UAVs sends a message to the destination UAV. For example, the flight command (e.g., waypoint update) and the position information of the N UAVs are exchanged every time slot to ensure safe UAV operations and sufficiently long UAV distance.

At the m -th of the N subslots in time slot k , UAV m at location $\mathbf{L}_m^{(k)} \in \mathbf{R}^3$ that is at least $D_{\min}$ away from the nearest neighboring UAV is moving at speed $v_m^{(k)}$ and sends $c_m^{(k)}$ -byte message to UAV $d_m^{(k)}$ with power $p_m^{(k)} \in [P_{\min}, P_U]$. Before the transmission, UAV m measures the battery level $b_m^{(k)}$, evaluates the number of the neighboring UAVs $\beta_m^{(k)}$, estimates the channel gain to the other UAVs $\mathbf{h}_m^{(k)} = \{h_{m,j}^{(k)} | 1 \leq j \neq m \leq N\}$, and chooses UAV $r_m^{(k)} \in \{j | 1 \leq j \neq m \leq N\}$ to relay the message using the decode-and-forward protocol with power $x_m^{(k)} \in [0, P_U]$.

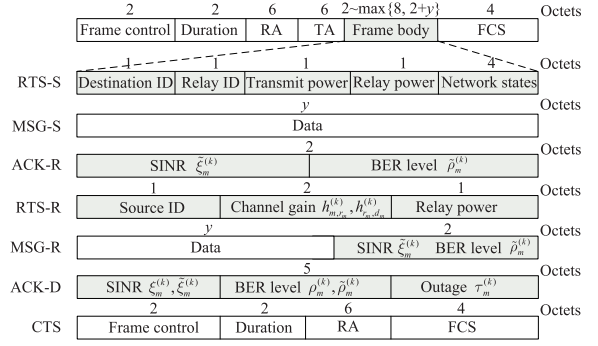


Fig. 2. Frame format of the swarm communication protocol, in which each UAV exchanges the network states and communication performance within the RTS/CTS/ACK messages.

B. Swarm Communication Protocol

As shown in Fig. 1, UAV m broadcasts the request to send message (RTS-S) that consists of destination ID $d_m^{(k)}$, selected relay ID $r_m^{(k)}$, transmit power $p_m^{(k)}$, relay power $x_m^{(k)}$ and network states $\Upsilon_m^{(k)}$ (e.g., channel gain $\mathbf{h}_m^{(k)}$). Upon receiving the RTS-S, UAV $r_m^{(k)}$ sends CTS to avoid the hidden node problem. UAV m then sends the $c_m^{(k)}$ -byte message (MSG-S) to UAV $r_m^{(k)}$ with transmit power $p_m^{(k)}$.

UAV $r_m^{(k)}$ evaluates $\text{SINR } \xi_m^{(k)}$, decodes the message and calculates the BER of the recent B received messages denoted by $\tilde{\rho}_m^{(k)}$. To reduce the communication overhead of the feedback channel, the BER is quantized into L levels with a ceiling function, i.e., $\tilde{\rho}_m^{(k)} = \lceil -\ln(\tilde{\rho}_m^{(k)}) \rceil$, in the acknowledgment message ACK-R. UAV $r_m^{(k)}$ then broadcasts the RTS-R message consists of source ID m , the estimated channel states of the source-relay link $\mathbf{h}_{m,r_m}^{(k)}$ and the relay-destination link $\mathbf{h}_{r_m,d_m}^{(k)}$ and relay power $x_m^{(k)}$. Upon receiving the CTS message, UAV $r_m^{(k)}$ sends the MSG-R that consists of the message in MSG-S, $\text{SINR } \xi_m^{(k)}$ and BER level $\tilde{\rho}_m^{(k)}$ with power $x_m^{(k)}$.

If successfully decoding the relay message, UAV $d_m^{(k)}$ evaluates the signal $\text{SINR } \xi_m^{(k)}$ and estimates the BER level $\rho_m^{(k)}$ of the B recent received messages. If no relay or source message is received by the destination, the communication outage indicator $\tau_m^{(k)}$ is set as 1, and 0 otherwise. UAV $d_m^{(k)}$ then broadcasts the feedback information including the signal $\text{SINR } \xi_m^{(k)}$, the message BER level $\rho_m^{(k)}$ and the communication outage indicator $\tau_m^{(k)}$ to all UAVs in the ACK-D message.

UAV m obtains message BER level $\rho_m^{(k)}$ and communication outage indicator $\tau_m^{(k)}$ from the ACK-D message and evaluates energy consumption $w_m^{(k)}$ in the previous time slot. The other UAV $n \in \{j | j \neq m, r_m^{(k)}, d_m^{(k)}\}$ keeps silent and updates the learning parameters based on the learning experience shared by UAV m from the RTS/CTS/ACK messages.

As an example of the frame format as shown in Fig. 2, the shared network states $\Upsilon_m^{(k)}$ with 4 octets in the RTS-S consist of the channel gain $\mathbf{h}_m^{(k)}$ (28 bits) and the number of the neighboring UAVs $\beta_m^{(k)}$ (4 bits). The BER is quantized into $L = 15$ levels with at most 6 bits. The signal $\text{SINR } \xi_m^{(k)}$ occupies up to 10 bits in the ACK-D message. This work is applicable to the other frame formats.

C. Jamming Model

A jammer located at $\mathbf{L}_j^{(k)}$ is assumed to track the UAV swarm and sends jamming signals to block the swarm communications. In particular, a reactive jammer as given in [29] senses the ongoing swarm transmission with signal detection techniques (e.g., energy detection) and estimates the current swarm transmission state, i.e., whether the UAV in the swarm is transmitting data or staying idle. Once detecting the activities on the target channel, the jammer sends jamming signals to decrease the signal SINR and increase the message BER received at the destination.

The jammer can adjust the jamming policy based on the estimated swarm transmission state to improve the energy efficiency. For example, the jammer observes the previous swarm transmission state $\varrho \in \{0, 1\}$, i.e., 1 indicates a successful swarm transmission (a failed jamming) and 0 otherwise. RL algorithms such as Q-learning can be applied to choose the jamming power $z \in \{jP_j/H | 0 \leq j \leq H\}$ based on the estimated transmission state to maximize the utility u_j that depends on the previous swarm transmission state and the jamming cost.

D. Channel Model

The channel state between UAV m and UAV n with distance l denoted by $h_{m,n}^{(k)}$ is assumed to be quasi-static during the time slot and depends on the path loss model given by [30] as

$$PL(l) = \nu 10 \log_{10} l + PL_0, \quad l \geq 1 \text{ m}, \quad (1)$$

where $\nu = 2.05$ is the path loss exponent and PL_0 is the reference path loss. The jammer locates at the ground and the elevation angle with UAV m is ϑ_m . According to [10], the channel state from UAV m to the jammer denoted by g_m is assumed to depend on the LoS link with probability

$$P_L = \left(1 + \varsigma_1 \exp\left(\varsigma_2 (\vartheta_m - \varsigma_1)\right) \right)^{-1} \quad (2)$$

and the non-LoS link with probability $1 - P_L$, where $\varsigma_1 = 11.95$ and $\varsigma_2 = 0.14$ are constant parameters specified by the environment.

E. Problem Formulation

The relay node $r_m^{(k)}$, transmit power $p_m^{(k)}$ and relay power $x_m^{(k)}$ are chosen to maximize the weighted sum of the BER level $\rho_m^{(k)}$, the energy consumption $w_m^{(k)}$ and the communication outage indicator $\tau_m^{(k)}$, given by

$$\max_{r_m^{(k)}, p_m^{(k)}, x_m^{(k)}} \rho_m^{(k)} \left(r_m^{(k)}, p_m^{(k)}, x_m^{(k)} \right) - c_1 w_m^{(k)} \left(r_m^{(k)}, p_m^{(k)}, x_m^{(k)} \right) - c_2 \tau_m^{(k)} \left(r_m^{(k)}, p_m^{(k)}, x_m^{(k)} \right), \quad (3a)$$

$$\text{s.t.} \quad P_{\min} \leq p_m^{(k)} \leq P_U, \quad (3b)$$

$$0 \leq x_m^{(k)} \leq P_U, \quad (3c)$$

$$1 \leq r_m^{(k)} \neq m \leq N. \quad (3d)$$

The optimal relay selection and power allocation $\mathbf{a}_m^{(k)} = \{r_m^{(k)}, p_m^{(k)}, x_m^{(k)}\}$ in (3) depends on the received jamming

power and the channel gain of the relay-destination links, which are rarely known accurately in time by UAV m . This problem can be addressed by RL since the repeated anti-jamming communication policy decision process of UAV can be viewed as a Markov decision process, in which the future state of the UAV is independent of the previous state and communication policy if given the current state and policy. Time index k and UAV ID m are omitted if no confusion occurs. The notations adopted in this paper are listed in Table I for convenience of reference.

IV. MULTI-AGENT RL-BASED UAV SWARM ANTI-JAMMING COMMUNICATIONS

We propose a multi-agent RL-based UAV swarm communication scheme (MURP) to choose the relay node, transmit power and relay power against jamming. Based on the observed network topology, the channel states, previous communication performance and the network states shared by the neighboring UAVs, the communication policy is chosen based on the policy distribution π that introduces the randomness to improve the exploration efficiency. The average policy distribution $\bar{\pi}$ and the Q-values Q guide the update of π based on the policy learning mechanism to stabilize the swarm learning process. Based on transfer learning, the shared UAV communication experiences are exploited to initialize the learning parameters such as Q-values and policy distribution for efficient exploration.

A. State

At subslot m in time slot k , UAV m evaluates the data size c and the number of UAVs within its communication range β , measures the battery level b and the channel gain with other UAVs denoted by $\mathbf{h}_m$, and obtains previous communication performance from the ACK message, including the message BER level ρ , the signal SINR ξ , the communication indicator τ , and the shared states of neighboring UAVs Ψ_{-m} from the RTS messages.

The state $\mathbf{s}^{(k)}$ consists of the number of neighboring UAVs β , the battery level b , channel states with neighboring UAVs $\mathbf{h}_m$, previous communication performance obtained from the feedback, i.e., the message BER level ρ , the signal SINR ξ and the communication indicator τ , and the shared network states Ψ_{-m} . Thus, the state is given as

$$\mathbf{s}^{(k)} = [\rho, \xi, \mathbf{h}_m, \beta, b, \tau, \Psi_{-m}]. \quad (4)$$

B. Action

The anti-jamming communication policy $\mathbf{a}^{(k)} = [r^{(k)}, p^{(k)}, x^{(k)}] \in \mathbf{A}$, including the relay UAV $r^{(k)}$, the transmit power $p^{(k)}$ and the relay power $x^{(k)}$. At time slot k , UAV m transmits the message with power

$$p^{(k)} \in \left\{ \frac{P_U - P_{\min}}{M - 1} j + P_{\min} \mid 0 \leq j \leq M - 1 \right\} \quad (5)$$

and chooses UAV $r^{(k)}$ to relay the message to destination with power $x^{(k)} \in \{jP_U/(M - 1) | 0 \leq j \leq M - 1\}$ or directly

difference between current expected Q-value and previous expected Q-value given the parameter ϖ_1 and ϖ_2 , i.e.,

$$\delta(s^{(k)}) = \varpi_1 + \frac{\varpi_2}{1 + \exp(\mathbb{E}_{a \sim \pi}[Q] - \mathbb{E}_{a \sim \bar{\pi}}[Q])}. \quad (11)$$

A lower $\delta(s^{(k)})$ results in more stable learning for the swarm with rapidly changing channel states and large swarm size. The maximum and minimum policy learning rate can be set adaptively according to the size of action set and the number of UAVs in the swarm. For example, $\varpi_1 = 0.002$ and $\varpi_2 = 0.018$ have been shown to enable stable learning for the 10-UAV swarm each with 5 power levels and 6 available relays.

For each communication policy $a \in \mathbf{A}$, the probability $\pi(s^{(k)}, a)$ is updated with an update value $\alpha(s^{(k)}, a)$ that depends on both the Q-value $Q(s^{(k)}, a)$ and the policy learning rate $\delta(s^{(k)})$. Specifically, the policy with maximum Q-value, i.e., $\tilde{a} = \arg\max_x Q(s^{(k)}, x)$, will have a positive update value and the other policies have negative update values given by (12), as shown at the bottom of the next page.

The policy distribution $\pi(s^{(k)}, \cdot)$ is updated by

$$\pi(s^{(k)}, \cdot) \leftarrow \pi(s^{(k)}, \cdot) + \alpha(s^{(k)}, \cdot). \quad (13)$$

The network states that consist of the channel states h_{-m} , the number of neighboring UAVs β_{-m} and the BER level ρ_{-m} shared in the RTS/CTS/ACK messages sent by the κ nearest UAVs are used to formulate the learning state Ψ_{-m} in (4) to enhance the perceptive field and accelerate learning. Based on the feedback in the ACK messages of β_m neighboring UAV $j \in \mathcal{W}_m$ and the power allocation in the RTS/CTS messages, the utility $u_j^{(k)}$ is evaluated via (6). The shared learning experience of UAV j includes the state $s_j^{(k)}$, the anti-jamming communication policy $a_j^{(k)}$, the utility $u_j^{(k)}$ and the next state $s_j^{(k+1)}$, i.e., $\psi_j^{(k)} = (s_j^{(k)}, a_j^{(k)}, u_j^{(k)}, s_j^{(k+1)})$.

The experience of UAV m , i.e., $\psi_m^{(k)} = (s_m^{(k)}, a_m^{(k)}, u_m^{(k)}, s_m^{(k+1)})$ and the shared experiences of neighboring UAVs $\mathcal{W}_m$, i.e., $\Phi_{-m}^{(k)} = \{\psi_j^{(k)} | j \in \mathcal{W}_m\}$ are stored in the experience pool Φ_m and J experiences are randomly selected from Φ_m to update the Q-values, the average policy distribution and the policy distribution to further improve the sample efficiency and accelerate the policy optimization, as summarized in **Algorithm 1**.

V. DEEP MULTI-AGENT RL-BASED UAV SWARM ANTI-JAMMING COMMUNICATIONS

We further propose a deep multi-agent RL-based UAV swarm relay selection and power allocation scheme (DMURP) for the swarm to accelerate the learning process and mitigate the state quantization error under the dynamic swarm with high moving speed. This scheme designs three deep neural networks including the policy network, the average policy network and the Q-value network to approximate the policy distribution, the average policy distribution and the Q-values of state-action pairs. The Q-value network and the average policy network both act as the evaluator to assess the policy distribution and guide the update of the policy network. The

Algorithm 2 Deep Multi-Agent RL-Based UAV Swarm Anti-Jamming Communications

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1:  $\theta_{Q/P/A} = \sum_{i=1}^W \hat{\theta}_{Q/P/A}^i, \mathcal{D} = \emptyset$ 
2: Initialize  $\rho_m^{(0)}, \tau_m^{(0)}, \Psi_{-m}^{(0)}, \lambda, \gamma, \phi, \varpi_1$  and  $\varpi_2$ 
3: for  $k = 1, 2, \dots, K$  do
4:   for  $m = 1, 2, \dots, N$  do
5:     Formulate  $s_m^{(k)}$  via (4)
6:     Input  $s_m^{(k)}$  to the policy network and output
        $\pi_m(s_m^{(k)}, \cdot)$ 
7:     Choose  $a_m^{(k)} = \{r_m^{(k)}, p_m^{(k)}, x_m^{(k)}\} \in \mathbf{A}$  based on
        $\pi_m(s_m^{(k)}, \cdot)$ 
8:     Obtain  $\rho_m^{(k)}, \xi_m^{(k)}$  and  $\tau_m^{(k)}$  from the feedback
9:     Evaluate  $u_m^{(k)}$  via (6)
10:     $\mathcal{D} \leftarrow \mathcal{D} \cup \{e^{(k)}\}$ 
11:  end for
12:  for  $m = 1, 2, \dots, N$  do
13:     $\mathcal{D} \leftarrow \mathcal{D} \cup \Phi_{-m}^{(k)}$ 
14:    if  $|\mathcal{D}| \geq G$  then
15:      Randomly sample  $\mathcal{B}$  from  $\mathcal{D}$ 
16:      Update  $\theta_{Q/P/A}$  via (19)-(21)
17:    end if
18:  end for
19: end for

```

experience replay technique is applied to update the network weights and accelerate the convergence by exploiting the anti-jamming experiences and network states of neighboring UAVs, as summarized in **Algorithm 2**.

A. Learning Structure

The learning architecture of UAV m as shown in Fig. 4 has three neural networks, including the policy network (P-network) with weights θ_P , the Q-value network (Q-network) with weights θ_Q and the average policy network (A-network) with weights θ_A . With state $s^{(k)}$ as the input, P-network extracts the features of the state to estimate the policy distribution π . The policy $a^{(k)}$ is chosen based on the state $s^{(k)}$ and the output of the policy network, i.e., $\Pr(a^{(k)} = a) = \pi(s^{(k)}, a; \theta_P)$. Q-network and A-network both with the state $s^{(k)}$ as the input outputs Q-values $Q(s^{(k)}, \cdot)$ and average policy distribution $\bar{\pi}$, respectively. Similar to **Algorithm 1**, the network states $\Psi_{-m}^{(k)}$ and learning experiences $\Phi_{-m}^{(k)}$ shared by the neighboring UAVs are also used in the state formulation of UAV m and the update of neural network weights, respectively.

The Q-network and A-network serve as the evaluator to calculate the policy learning rate and guide the P-network update. Experience replay technique is applied to update the network weights by exploiting the learning experiences of neighboring UAVs. As summarized in **Algorithm 2**, the network weights of three networks are initialized with the weighted average weights obtained for the W neighboring UAVs, i.e., $\theta_{Q/P/A} = \sum_{i=1}^W \hat{\theta}_{Q/P/A}^i$.

B. Network Architecture

Three networks have the same architecture with an input layer, two hidden layers and an output layer, which are designed to learn the mapping relation between the state and network output. The input size Ω mainly depends on the number of neighboring UAVs and the size of observations extracted from other UAVs, which is given as $\Omega = 5 + \beta + |\Psi_{-m}|$. The output size depends on the size of action set, i.e., $|\mathbf{A}| = M^2(N - 1)$. According to [31], the number of neural nodes in two hidden layers f_1 and f_2 are determined based on the output size $|\mathbf{A}|$ and the learning samples K .

Specifically, the number of neural nodes in the second hidden layer depends on the output size and the learning samples, given as

$$f_2 = |\mathbf{A}| \sqrt{\frac{K}{|\mathbf{A}| + 2}}. \quad (14)$$

The number of neural nodes in the first hidden layer depends on the f_2 and the learning samples, given as

$$f_1 = \sqrt{(|\mathbf{A}| + 2)K} + 2\sqrt{\frac{K}{|\mathbf{A}| + 2}}. \quad (15)$$

The network weights of three networks $\theta_{Q/P/A}$ consist of Ωf_1 weights between the input layer and the first hidden layer, $(f_1 + 1)f_2$ weights between the first and second hidden layer and $M^2(f_2 + 1)(N - 1)$ weights between the second hidden layer and the output layer.

C. Network Update

Similar to Algorithm 1, UAV m evaluates the utility via (6) based on the feedback information and observes the next state $\mathbf{s}^{(k+1)}$. The experience $e^{(k)}$ including the current state and action, the utility, and the next state, i.e., $e^{(k)} = (\mathbf{s}^{(k)}, \mathbf{a}^{(k)}, u^{(k)}, \mathbf{s}^{(k+1)})$ and the anti-jamming communication experiences of the neighboring β_m UAVs $\Phi_{-m}^{(k)}$ extracted from the RTS/CTS/ACK messages are stored in memory pool $\mathcal{D}$ to further improve the learning efficiency. The minibatch $\mathcal{B}$ is formulated with G experiences randomly sampled from $\mathcal{D}$ given by

$$\mathcal{B} = \left\{ \left(\mathbf{s}^{(l(j))}, \mathbf{a}^{(l(j))}, u^{(l(j))}, \mathbf{s}^{(l(j)+1)} \right) \in \mathcal{D} \mid 1 \leq j \leq G \right\}. \quad (16)$$

The experience replay technique is applied to update the neural network weights by using the stochastic gradient descent algorithm based on the minibatch $\mathcal{B}$. According to the update process in the Algorithm 1, the target values of three networks are calculated to update the network weights

by minimizing the mean square error between the network outputs and the target values. Specifically, for each experience $\{\tilde{\mathbf{s}}, \tilde{\mathbf{a}}, \tilde{u}, \tilde{\mathbf{s}}\} \in \mathcal{B}$, the target value of Q-network is given by

$$Q^T(\tilde{\mathbf{s}}, \tilde{\mathbf{a}}; \theta_Q) = \tilde{u} + \gamma \max_{\mathbf{a} \in \mathbf{A}} Q(\tilde{\mathbf{s}}, \mathbf{a}; \theta_Q). \quad (17)$$

The output of P-network guides the update of the A-network by calculating the target value of average policy distribution with

$$\bar{\pi}^T(\tilde{\mathbf{s}}, \tilde{\mathbf{a}}; \theta_A) = (1 - \phi) \bar{\pi}(\tilde{\mathbf{s}}, \tilde{\mathbf{a}}; \theta_A) + \phi \pi(\tilde{\mathbf{s}}, \tilde{\mathbf{a}}; \theta_P), \quad (18)$$

where $\phi \in [0, 1]$ is the learning rate for the average policy network. The target value of the P-network is calculated with both the outputs of A-network and Q-network by calculating the update value $\alpha(\tilde{\mathbf{s}}, \cdot)$ via (12).

Specifically, the weights of the Q-network θ_Q are updated by minimizing the loss function given as

$$\mathcal{L}(\theta_Q) = \mathbb{E}_{\{\tilde{\mathbf{s}}, \tilde{\mathbf{a}}, \tilde{u}, \tilde{\mathbf{s}}\} \in \mathcal{B}} \left[\left(\tilde{u} + \gamma \max_{\mathbf{a} \in \mathbf{A}} Q(\tilde{\mathbf{s}}, \mathbf{a}; \theta_Q) - Q(\tilde{\mathbf{s}}, \tilde{\mathbf{a}}; \theta_Q) \right)^2 \right]. \quad (19)$$

The weights of the A-network θ_A are updated by minimizing the loss function that depends on the difference between the policy distribution and the average policy distribution, i.e.,

$$\mathcal{L}(\theta_A) = \mathbb{E}_{\{\tilde{\mathbf{s}}, \tilde{\mathbf{a}}, \tilde{u}, \tilde{\mathbf{s}}\} \in \mathcal{B}} \left[\phi^2 \sum_{\mathbf{a} \in \mathbf{A}} \left(\bar{\pi}(\tilde{\mathbf{s}}, \mathbf{a}; \theta_A) - \pi(\tilde{\mathbf{s}}, \mathbf{a}; \theta_P) \right)^2 \right]. \quad (20)$$

The weights of the P-network θ_P are updated by minimizing the loss function that depends on the policy update value, which is given by (21), as shown at the bottom of the next page.

VI. PERFORMANCE ANALYSIS

We formulate the swarm anti-jamming communication game and analyze the performance bound via the NE of the game including the messages BER, the swarm energy consumption and the utility. In addition, the computational complexities of our proposed UAV swarm anti-jamming communication schemes are investigated.

A. Game Model

In the swarm anti-jamming communication game between the swarm consisting of N UAVs $\mathcal{N} = \{1, 2, \dots, N\}$ and the jammer $\mathcal{J}$, the swarm chooses the relay node and power

$$\alpha(\mathbf{s}^{(k)}, \mathbf{a}) = \begin{cases} -\min \left\{ \pi(\mathbf{s}^{(k)}, \mathbf{a}), \frac{\exp(-Q(\mathbf{s}^{(k)}, \mathbf{a})) \delta(\mathbf{s}^{(k)})}{\sum_{\mathbf{x} \in \mathbf{A} \setminus \tilde{\mathbf{a}}} \exp(-Q(\mathbf{s}^{(k)}, \mathbf{x}))} \right\}, & \mathbf{a} \neq \arg\max_{\mathbf{x}} Q(\mathbf{s}^{(k)}, \mathbf{x}) \\ \sum_{\mathbf{x} \in \mathbf{A} \setminus \tilde{\mathbf{a}}} \alpha(\mathbf{s}^{(k)}, \mathbf{x}), & \mathbf{a} = \arg\max_{\mathbf{x}} Q(\mathbf{s}^{(k)}, \mathbf{x}) \end{cases} \quad (12)$$

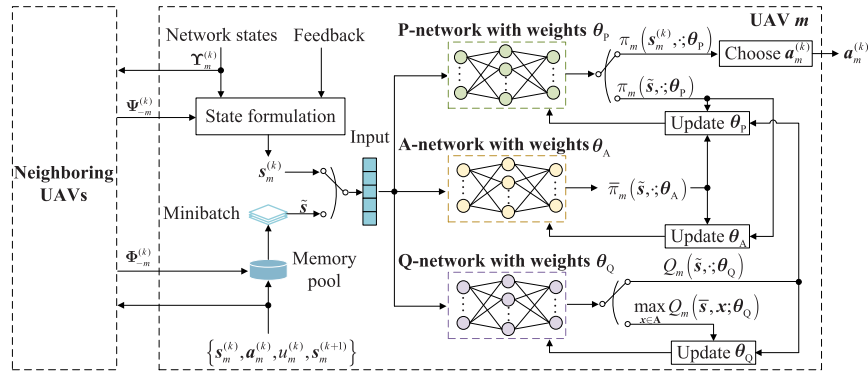


Fig. 4. Illustration of deep multi-agent RL-based UAV swarm relay selection and power allocation scheme, in which UAV m inputs $\mathbf{s}_m^{(k)}$ to the policy network for policy selection and switches to input the state of experiences in minibatch to three networks for network weights update.

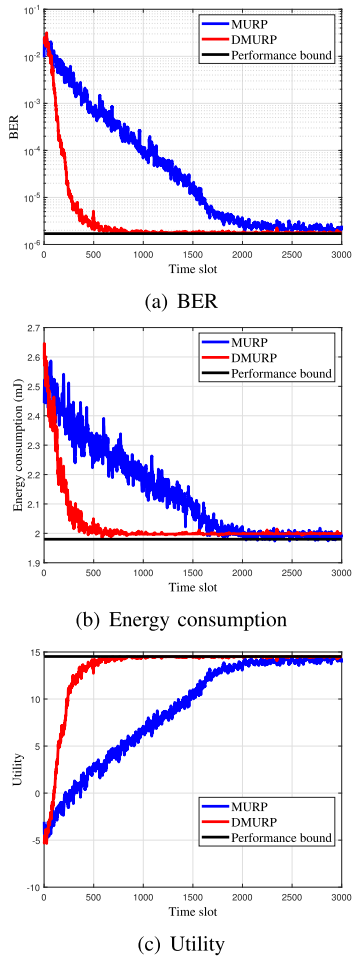


Fig. 5. Performance of the swarm anti-jamming communications with 5 UAVs and a random jammer.

allocation policy $\{[r_i, p_i, x_i], 1 \leq i \leq N\} \in \mathbf{A}^N$. UAV i transmits messages with power $p_i \in [P_{\min}, P_U]$ and selects UAV $r_i \in \mathcal{N}_{-i}$ to relay the messages to UAV $j \in \mathcal{N}_{-i}$ with power $x_i \in [0, P_U]$. The jammer chooses the power $z \in \mathbf{Z}$

to degrade the swarm communication performance with less jamming power.

The swarm aims to maximize the utility $\mathbf{u} = \{u_i, i \in \mathcal{N}\}$, in which the utility u_i of UAV i depends on the communication performance and energy consumption. The jamming utility u_J depends on both the swarm utility and the jamming power with coefficient ω_J , which is given as $u_J = -\sum_{i=1}^N u_i - \omega_J z$. Thus, the anti-jamming communication game model can be formulated by $\mathcal{G} = \{\{\mathcal{N}, \mathcal{J}\}, \{\mathbf{A}^N, \mathbf{Z}\}, \{\mathbf{u}, u_J\}\}$.

B. Performance Bound

Similar to [13], we derive the performance bound of the proposed schemes via analyzing the NE of the anti-jamming communication game between UAV i and the jammer. In the game $\mathcal{G}$, UAV $i \in \mathcal{N}$ aims to transmit messages with size c to UAV $j \in \mathcal{N}_{-i}$ and chooses relay node and power allocation policy $\mathbf{a} = [r, p, x] \in \mathbf{A}$ to maximize utility u_i given by

$$u_i = \text{sgn}(x) \min \left\{ \frac{ph_{i,r}}{\sigma^2 + zg_r}, \frac{xh_{r,j}}{\sigma^2 + zg_j} \right\} + (1 - \text{sgn}(x)) \frac{ph_{i,j}}{\sigma^2 + zg_j} - \omega_U(p + x), \quad (22)$$

where σ^2 is the receiver noise power, ω_U denotes the importance of energy consumption and $\text{sgn}(x)$ is the signum function, i.e., $\text{sgn}(x) = 1$ if $x > 0$ and $\text{sgn}(x) = 0$ if $x = 0$. The jammer chooses jamming power $z \in [0, P_J]$ to maximize utility $u_{J,i}$ given as

$$u_{J,i} = -u_i - \omega_J z. \quad (23)$$

Theorem 1: The performance bound of the game $\mathcal{G}$ for UAV $i \in \mathcal{N}$ to destination UAV $j \in \mathcal{N}_{-i}$ with QPSK is given by

$$\bar{\xi} = \frac{1}{2} \text{erfc} \left(\sqrt{\frac{P_U h_{i,j}}{2\sigma^2}} \right) \quad (24a)$$

$$\bar{C} = \omega_E P_U c \quad (24b)$$

$$\bar{u}_i = \frac{P_U h_{i,j}}{\sigma^2} - \omega_U P_U \quad (24c)$$

$$\mathcal{L}(\theta_P) = \mathbb{E}_{\{\tilde{s}, \tilde{a}, \tilde{u}, \tilde{s}\} \in \mathcal{B}} \left[\sum_{a \in \mathcal{A}} \left(\alpha(\tilde{s}, a; \pi(\tilde{s}, a; \theta_P), Q(\tilde{s}, \cdot; \theta_Q), \bar{\pi}(\tilde{s}, \cdot; \theta_A)) \right)^2 \right]. \quad (21)$$

if

$$\max_{r \in \mathcal{N} \setminus \{i,j\}} \left\{ h_{i,r}, h_{r,j}, \frac{h_{i,j} \omega_U P_U g_j}{\omega_J \sigma^2} \right\} < \omega_U \sigma^2 < h_{i,j}, \quad (25)$$

where ω_E is the unit transmission energy consumption.

Proof: See Appendix I. $\square$

Remark 1: UAV i chooses to transmit messages directly to the destination UAV j with maximum power P_U and the jammer keeps silent if the jamming cost is high and the channel condition of the source-destination link is better than the channel conditions of source-relay and relay-destination links as given in (25). Simulation results in Fig. 5 show that the proposed MURP and DMURP decrease the message BER and improve the UAV utility and the communication performance converges to the bound given by (24) after 500 and 2000 time slots, respectively.

Theorem 2: The performance bound of the game $\mathcal{G}$ for UAV $i \in \mathcal{N}$ to destination UAV $j \in \mathcal{N}_{-i}$ with QPSK is given by

$$\hat{\xi} = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{P_U \max_{r \in \mathcal{N}} h_{r,j}}{2(\sigma^2 + P_J g_j)}} \right) \quad (26a)$$

$$\hat{C} = \omega_E (P_U + P_{\min}) c \quad (26b)$$

$$\hat{u}_i = \frac{P_U \max_{r \in \mathcal{N}} h_{r,j}}{\sigma^2 + P_J g_j} - \omega_U (P_U + P_{\min}) \quad (26c)$$

if $\forall r \in \mathcal{N} \setminus \{i, j\}$,

$$h_{i,j} < \omega_U (\sigma^2 + P_J g_j), \quad (27)$$

$$\omega_U (\sigma^2 + P_J g_j) < h_{r,j} < \frac{P_{\min} (\sigma^2 + P_J g_j)}{P_U (\sigma^2 + P_J g_r)} h_{i,r}, \quad (28)$$

$$\max_{r \in \mathcal{N}} h_{r,j} > \max \left\{ \frac{\omega_J (\sigma^2 + P_J g_j)^2}{P_U g_j}, \frac{\omega_U (P_U + P_{\min}) (\sigma^2 + P_J g_j)}{P_U} \right\}. \quad (29)$$

Proof: See Appendix II. $\square$

Remark 2: UAV i transmits with minimal power $P_{\min}$ and chooses UAV $r^* = \arg \max_{r \in \mathcal{N}} h_{r,j}$ with power P_U to relay the message and the jammer chooses the maximum power P_J if the jamming cost is low and channel conditions between destination UAV j and UAV r^* is better than channel conditions with other UAVs as given in (27)-(29).

C. Computational Complexity

The computational complexities of the proposed MURP and DMURP mainly depend on the number of multiplications in the policy selection and the update of the Q-values, policy distribution, average policy distribution and three neural networks. Specifically, the computational complexity of MURP in the policy selection is $O(M^2 N)$. In the policy distribution learning process, MURP samples J experiences per time slot to update the Q-values, the policy distribution and the average policy distribution and the resulting computational complexity is $O(JM^2 N)$.

In the proposed DMURP, the number of multiplications depends on the forward propagation of policy selection and back propagation of weights update in the Q-network, policy

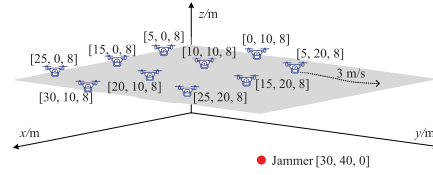


Fig. 6. Simulation setting of the UAV swarm communication against a jammer that randomly chooses jamming power from $\{5j | 0 \leq j \leq 6\}$ mW, in which each UAV sends 2-KB data aided by the other UAVs with transmit and relay power choosing from $\{20j | 1 \leq j \leq 5\}$ mW.

network and average policy network. Three networks share same network architecture and the input size, the number of neural nodes of two hidden layers, and the output size is Ω , f_1 , f_2 and $M^2(N-1)$, respectively. The size of two hidden layers given in (14) and (15) depend on the size of the action set and the number of learning samples K .

In the policy selection process, the policy network that performs forward propagation has l_F multiplications given as

$$l_F = \Omega f_1 + (f_1 + 1) f_2 + M^2 (f_2 + 1) (N - 1). \quad (30)$$

In the network weights update process, the policy network, average policy network and Q network performs l_F , l_F and $2l_F$ multiplications in the forward propagation, respectively. The policy network and average policy network both perform l_P multiplications in the back propagation given as

$$l_P = 2\Omega f_1 + 2(f_1 + 1) f_2 + 3M^2 (f_2 + 1) (N - 1). \quad (31)$$

The Q network performs l_Q multiplications in the back propagation given as

$$l_Q = 2\Omega f_1 + 2(f_1 + 1) f_2 + 3(f_2 + 1). \quad (32)$$

DMURP samples G experiences from the memory pool per time slot to update three network weights and the total number of multiplications in the policy selection and network weights update is

$$l_T = (4G + 1) l_F + 2l_P + l_Q. \quad (33)$$

Theorem 3: The computational complexity of the proposed MURP and DMURP is given by $O(JM^2 N)$ and $O(GKM^2 N)$, respectively.

Proof: See Appendix III. $\square$

Remark 3: The computational complexity of the proposed MURP mainly depends on the quantized power level M and the number of UAVs N and the number of selected experiences J for the policy learning. The computational complexity of DMURP increases with the batch size G , the number of UAVs N , the quantized power level M and the number of learning samples K . Therefore, the swarm has to improve the anti-jamming communication policy optimization efficiency and make a trade-off between the sample complexity and learning speed.

VII. SIMULATION RESULTS

As shown in Fig. 6, each of the 10 UAVs moves at 3 m/s and exchanges 2-KB messages every 500 ms at 2.4 GHz with up to 100 mW transmit power (i.e., $p, x \in \{20j | 1 \leq j \leq 5\}$

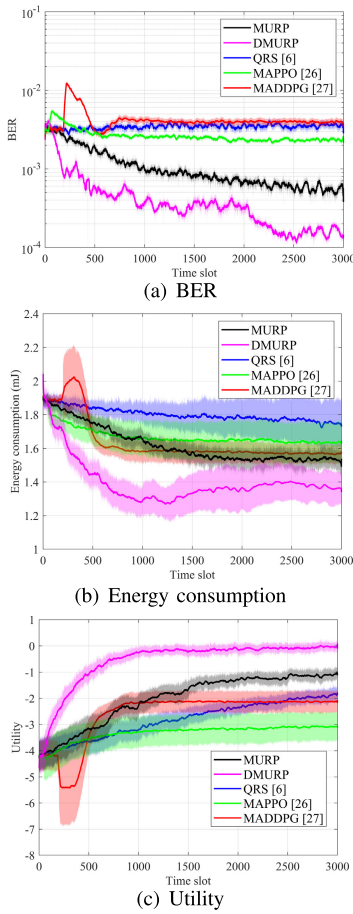


Fig. 7. Performance of the swarm communications with 10 UAVs and a jammer as shown in Fig. 6. The solid lines are average results with 10 UAVs and the light color areas are the error bars.

mW) against Gaussian jamming signals with power randomly chosen from $\{5j|0 \leq j \leq 6\}$ mW. In the simulations, the noise power is -110 dBm, the learning rate of neural networks is 0.001, $f_1 = f_2 = 128$, $J = 3$, $\lambda = 0.5$, $\gamma = 0.7$, $G = 64$, $\underline{\delta} = 0.002$ and $\bar{\delta} = 0.02$.

The performance gains of our proposed schemes over the benchmark QRS in [6], MAPPO in [26] and MADDPG in [27] are given in Fig. 7. For example, our proposed MURP exceeds QRS with 83.3% lower BER and 13.1% less energy consumption after 3000 time slots, due to the randomness that MURP introduces in the policy selection for better exploration. In addition, MURP reduces 53.3% BER and 8.1% energy consumption after 2000 time slots compared with MAPPO, due to the experience sharing among the neighboring UAVs.

DMURP further improves the performance with BER reduced from 7×10^{-4} to 2×10^{-4} and 11.2% less energy consumption after 2000 time slots and saves 50% learning time due to the deep neural networks that compress the state space and mitigate the state quantization error. In addition, DMURP outperforms MADDPG with 90.0% less BER and 19.0% less energy consumption, because that the policy selection mitigates the quantization error of MADDPG for the relay selection.

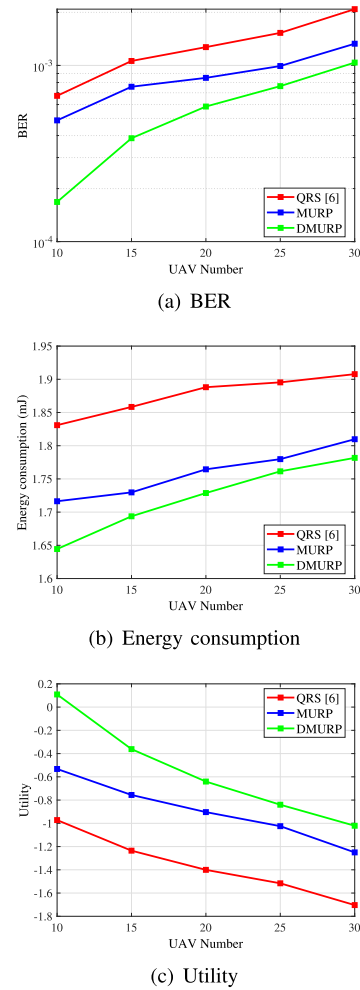


Fig. 8. Average performance of the swarm communication scheme over the UAVs and 20 runs each containing 1000 time slots with the swarm size increasing from 10 to 30.

The average performance as shown in Fig. 8 verifies the performance gain of our proposed schemes for the swarm ranging between 10 and 30 UAVs. For example, compared with QRS, MURP decreases 28.4% BER and the performance gain increases with the swarm size to 36.1%, due to the adaptive policy learning rate that results in stable learning. DMURP further reduces 23.0% BER and saves energy consumption by more than 0.02 mJ, because that the deep neural networks accelerate learning under the large swarm size.

VIII. EXPERIMENTAL RESULTS

In the experiments with the topology as shown in Fig. 9, each of the 4 UAVs equipped with Raspberry Pi sends 2-KB messages to the destination UAV with up to 100 mW transmit power at 2.4 GHz according to IEEE 802.11ac against a random jammer that chooses the jamming power from $\{5j|0 \leq j \leq 6\}$ mW.

As shown in Fig. 10, our proposed MURP consumes about 4000 time slots, i.e., 78.4 s, to improve the swarm communication performance, with the BER reduced from 4×10^{-4} to 5×10^{-6} and 10.6% energy consumption saving. DMURP further improves the performance with 12.5% less energy

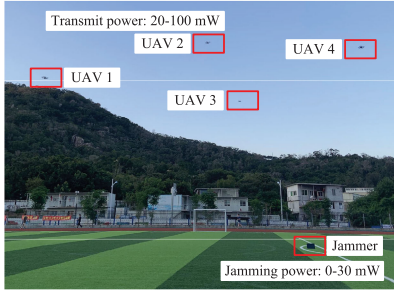
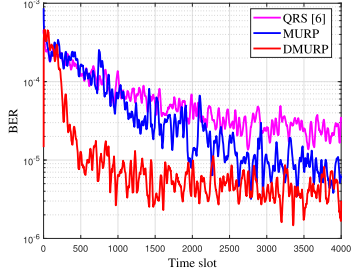
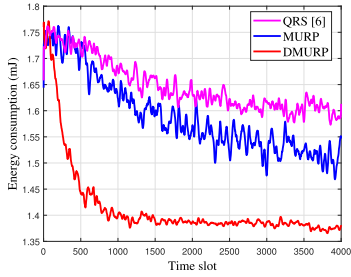


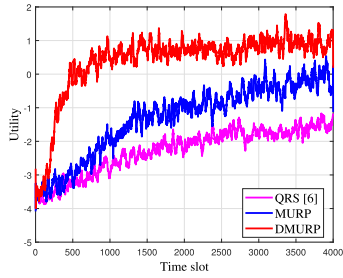
Fig. 9. Experimental settings of a 4-UAV swarm communication against a jammer choosing jamming power from $\{5^j | 0 \leq j \leq 6\}$ mW, in which each UAV sends messages with up to 100 mW transmit power at 2.4 GHz.



(a) BER



(b) UAV energy consumption



(c) Utility

Fig. 10. Average performance of the swarm communications with 4 UAVs and a jammer as shown in Fig. 9.

consumption and saves 62.5% learning time. Experimental results also verify the performance gain over the benchmark scheme QRS in [6]. For example, DMURP decreases 92.5% BER and 16.9% energy consumption after 1500 time slots (i.e., 41.4 s) compared with QRS.

IX. CONCLUSION

In this paper, we proposed a multi-agent RL-based swarm relay selection and power allocation scheme, which exploits the communication experiences and observations shared by the

neighboring UAVs to improve anti-jamming swarm communication performance. A deep RL version has been proposed to accelerate learning and further enhance the performance of highly dynamic swarms. Both simulation and experimental results verify the performance gain over the benchmark QRS, MAPPO and MADDPG. For example, in the experiments based on 4 UAVs against a jammer with up to 30 mW jamming power, our proposed scheme reduces the BER from 4×10^{-5} to 3×10^{-6} and saves 16.9% swarm energy consumption compared with QRS.

APPENDIX I PROOF OF THEOREM 1

Proof: By (22), we have

$$u_i([r, p, x], 0) = \text{sgn}(x) \min \left\{ \frac{ph_{i,r}}{\sigma^2}, \frac{xh_{r,j}}{\sigma^2} \right\} + (1 - \text{sgn}(x)) \frac{ph_{i,j}}{\sigma^2} - \omega_U(p + x). \quad (34)$$

If (25) holds, $\forall r \in \mathcal{N}_{-i}$ and $x > 0$, we have

$$u_i([r, p, x], 0) < \frac{ph_{i,j}}{\sigma^2} - \omega_U p = u_i([r, p, 0], 0). \quad (35)$$

For $x > 0$, we have

$$u_i([r, p, x], 0) = \min \left\{ \frac{ph_{i,r}}{\sigma^2}, \frac{xh_{r,j}}{\sigma^2} \right\} - \omega_U(p + x) \leq \frac{ph_{i,r}}{\sigma^2} - \omega_U(p + x). \quad (36)$$

If (25) holds, we have

$$\frac{\partial u_i([r, p, 0], 0)}{\partial p} = \frac{h_{i,j}}{\sigma^2} - \omega_U > 0, \forall p \in [P_{\min}, P_U]. \quad (37)$$

By (37), we have $u_i([r, p, 0], 0) \leq u_i([r, P_U, 0], 0)$. Thus, by (35), $\forall r \in \mathcal{N}_{-i}$, $p \in [P_{\min}, P_U]$ and $x \in [0, P_U]$, we have

$$u_i([r, p, x], 0) < u_i([r, P_U, 0], 0). \quad (38)$$

By (22) and (23), we have

$$u_{J,i}([r, P_U, 0], z) = -\frac{P_U h_{i,j}}{\sigma^2 + zg_j} + \omega_U P_U - \omega_J z. \quad (39)$$

If (25) holds, by (39), $\forall z \in [0, P_J]$, we have

$$\begin{aligned} \frac{\partial u_{J,i}([r, P_U, 0], z)}{\partial z} &= \frac{P_U h_{i,j} g_j}{(\sigma^2 + zg_j)^2} - \omega_J \leq \frac{P_U h_{i,j} g_j}{\sigma^4} - \omega_J < 0. \end{aligned} \quad (40)$$

Thus, we have $u_{J,i}([r, P_U, 0], z) \leq u_{J,i}([r, P_U, 0], 0)$. Thus, by (38) and (40), $([r, P_U, 0], 0)$ is an NE of $\mathcal{G}$ and the resulting performance bound is given by (24).

APPENDIX II PROOF OF THEOREM 2

Proof: By (22), $\forall r \in \mathcal{N}_{-i}$ and $x > 0$, we have

$$u_i([r, p, x], P_j) = \min \left\{ \frac{ph_{i,r}}{\sigma^2 + P_j g_r}, \frac{xh_{r,j}}{\sigma^2 + P_j g_j} \right\} - \omega_U(p + x). \quad (41)$$

If (28) holds, we have

$$\frac{xh_{r,j}}{\sigma^2 + P_j g_j} \leq \frac{P_U h_{r,j}}{\sigma^2 + P_j g_j} < \frac{P_{\min} h_{i,r}}{\sigma^2 + P_j g_r} \leq \frac{ph_{i,r}}{\sigma^2 + P_j g_r}. \quad (42)$$

Thus, by (41) and (42), $\forall r \in \mathcal{N}_{-i}$ and $x \neq 0$

$$u_i([r, p, x], P_j) = \frac{xh_{r,j}}{\sigma^2 + P_j g_j} - \omega_U(p + x). \quad (43)$$

If (28) holds, we have

$$\frac{\partial u_i([r, p, x], P_j)}{\partial x} = \frac{h_{r,j}}{\sigma^2 + P_j g_j} - \omega_U > 0 \quad (44)$$

and

$$\frac{\partial u_i([r, p, x], P_j)}{\partial p} = -\omega_U < 0. \quad (45)$$

Thus, by (44) and (45),

$$u_i([r, p, x], P_j) \leq u_i([r, P_{\min}, P_U], P_j). \quad (46)$$

For $x \neq 0$ and $\forall r \in \mathcal{N}_{-i}$, we have

$$u_i([r, P_{\min}, P_U], P_j) \leq u_i([r^*, P_{\min}, P_U], P_j), \quad (47)$$

if

$$r^* = \underset{r \in \mathcal{N} \setminus \{i,j\}}{\operatorname{argmax}} h_{r,j}. \quad (48)$$

By (22), if (27) and (29) hold,

$$u_i([r, p, 0], P_j) = \frac{ph_{i,j}}{\sigma^2 + P_j g_j} - \omega_U p < 0 \quad (49)$$

and

$$u_i([r^*, P_{\min}, P_U], P_j) > 0. \quad (50)$$

By (49) and (50), $\forall r \in \mathcal{N}_{-i}$, $p \in [P_{\min}, P_U]$ and $x \in [0, P_U]$, we have

$$u_i([r, p, x], P_j) \leq u_i([r^*, P_{\min}, P_U], P_j). \quad (51)$$

By (22) and (23), we have

$$u_{j,i}([r^*, P_{\min}, P_U], z) = -\frac{P_U h_{r^*,j}}{\sigma^2 + zg_j} + \omega_U(P_U + P_{\min}) - \omega_J z. \quad (52)$$

If (29) holds, $\forall z \in [0, P_j]$, we have

$$\begin{aligned} \frac{\partial u_{j,i}([r^*, P_{\min}, P_U], z)}{\partial z} &= \frac{P_U h_{r^*,j} g_j}{(\sigma^2 + zg_j)^2} - \omega_J \geq \frac{P_U h_{r^*,j} g_j}{(\sigma^2 + P_j g_j)^2} - \omega_J > 0. \end{aligned} \quad (53)$$

Thus, we have

$$u_{j,i}([r^*, P_{\min}, P_U], z) \leq u_{j,i}([r^*, P_{\min}, P_U], P_j). \quad (54)$$

By (51) and (54), $([r^*, P_{\min}, P_U], P_j)$ is an NE of $\mathcal{G}$ and the resulting performance bound is given by (26).

APPENDIX III PROOF OF THEOREM 3

Proof: By (30)-(33), the computational complexity of DMURP is given by

$$\Upsilon = \mathcal{O}((4G + 7)\Omega f_1 + (4G + 7)(f_1 + 1)f_2 + M^2(N - 1)(4G + 7)(f_2 + 1) + 3f_2) \quad (55)$$

$$= \mathcal{O}(G\Omega f_1 + Gf_1 f_2 + GM^2 N f_2 + GM^2 N). \quad (56)$$

As $K \gg M^2 N$ in [31], by (14), (15) and (56), we have

$$\begin{aligned} \Upsilon &= \mathcal{O}\left(G\Omega\sqrt{(M^2 N + 2)K} + GM^4 N^2 \sqrt{\frac{K}{M^2 N + 2}} \right. \\ &\quad \left. + GKM^2 N + 2K + GM^2 N\right) \end{aligned} \quad (57)$$

$$= \mathcal{O}(GKM^2 N). \quad (58)$$

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